

9 Many factors affect the rate of cellular respiration.

- (a)** Pyruvate carboxylase (PC) deficiency is a rare disease. The rate of respiration in the cells of a person with PC deficiency is much lower than normal.

The passage describes the occurrence of PC deficiency in human populations.

Complete the passage by using the most appropriate scientific terms.

PC deficiency affects one in 250 000 people. A person either has the disease or does not, so there are no intermediate forms of the disease. PC deficiency is caused by a genetic mutation in one gene. Whether a person develops the disease is not influenced by factors. The distribution of PC deficiency in human populations shows variation. [2]

- (b)** Coenzymes are important in aerobic respiration.

Outline the roles of named coenzymes in aerobic respiration.

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[Total: 7]